Yen Giang Nguyen

Van Andel Institute | Department of Neurodegenerative Science
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EDUCATION

B.A., Biology & Mathematics, *magna cum laude* 2023 - 2026
Kalamazoo College, Kalamazoo, MI

- **GPA:** 3.842 | Dean's List
- **Concentration:** Neuroscience & Biological Physics
- **Coursework:** Linear Algebra, Multivariable Calculus, Mathematical Statistics, Topology & Real Analysis

RESEARCH EXPERIENCE

Research Data Technician Jul 2026 - Now
Henderson Laboratory at Van Andel Institute, Grand Rapids, MI

- Computational analysis and modeling of neurodegenerative disease pathology and progression for VAI Brain Bank.

Bioinformatics Intern Jun 2024 - Jun 2026
Henderson Laboratory at Van Andel Institute, Grand Rapids, MI

❖ **Project 1: CosMx vs. Xenium Comparative Analysis**

- Developed a workflow for comparing the performance of the 1000-plex CosMx Mouse Neuro Panel to the 247-plex Xenium Mouse Neuro using R on high-performance computing (HPC) cluster.
- Annotated cell types in both datasets using supervised cell typing with the InSituType package, presenting results through flightpath plots, UMAP, and heatmaps.
- Conducted differential expression analysis of CosMx and Xenium through Seurat and MAST packages and visualized comparative results through volcano and violin plots.
- Findings contributed to Van Andel Institute's decision to acquire an in-house Xenium platform.

❖ **Project 2: VAI Brain Bank Analysis**

- Analyzed quantitative pathology data from 200 postmortem human brains in the VAI Brain Bank, focusing on hyperphosphorylated tau, amyloid beta (Aβ), α-synuclein, and TDP-43 accumulation across multiple brain regions using Python.
- Developed a pseudo-longitudinal staging framework using dose-response curve to align patients along disease progression trajectories for each brain region, whose kinetics can be mathematically fitted and explored.
- Performed dimensionality reduction and disease subtyping through principal component analysis (PCA), identifying distinct disease groupings such as Alzheimer's disease and progressive supranuclear palsy in tau pathology.
- Applied the Subtype and Stage Inference (SuStaIn) framework to model heterogeneous disease subtypes and progression stages from cross-sectional neuropathology data.

HONORS AND AWARDS

- **Undergraduate Travel Grant, Computational and Systems Neuroscience** 2026
  Competitive registration, travel, and lodging support for undergraduate attendance at COSYNE.
- **The Clarke Benedict Williams Prize, Kalamazoo College** 2026
  Highest graduating class distinction for achievement in mathematics and allied sciences.
- **Senior Integrated Project Honors, Department of Biology, Kalamazoo College** 2025
  Awarded to selected outstanding senior research projects in biology.
- **McColl Endowment Travel Award, Department of Biology, Kalamazoo College** 2025
- **Department of Biology Travel Award, Kalamazoo College** 2024
  Travel support to attend Neuroscience 2024.
- **Lux Esto Merit Scholarship, Kalamazoo College** 2023
  Four-year merit scholarship awarded for academic achievement.
- **CCIR Spotlight Scholar, Cambridge Centre for International Research** 2022
  Selected for distinction in an independently completed research project.

PRESENTATIONS

❖ **Oral Presentation**

- **Quantitative Brain Banking Offers New Insights into Neurodegenerative Disease Progression**

Nguyen YG, Cook J, Noe C, Abe K, Turner L, Johansson A, Rohrer D, Williams E, Moore D, Flanagan M, Henderson MX. *Grand Challenges in Parkinson's Disease.* 2026

- **Comparing Single-Cell Spatial Transcriptomics Techniques**

Nguyen YG, Goralski T, Breton L, Meyerdirk L, Adams M, Henderson MX. *Aligning Science Across Parkinson's – Team Lee's Monthly Meeting.* 2025

- **Comparing Single-Cell Spatial Transcriptomics Techniques**

Nguyen YG, Goralski T, Breton L, Meyerdirk L, Adams M, Henderson MX. *Aligning Science Across Parkinson's - Spatial Transcriptomics Meeting.* 2024

❖ **Poster Presentation**

- **Quantitative Brain Banking Offers New Insights into Neurodegenerative Disease Progression**

Nguyen YG, Cook J, Noe C, Abe K, Turner L, Johansson A, Rohrer D, Williams E, Moore D, Flanagan M, Henderson MX. *Grand Challenges in Parkinson's Disease.* 2026

- **Gene Therapy in the Central Nervous System: ImageJ as a Tool for Image Analysis**

Nguyen YG. *Neuroscience 2024, Society for Neuroscience.* 2024

- **Comparing Single-Cell Spatial Transcriptomics Techniques**

Nguyen YG, Goralski T, Breton L, Meyerdirk L, Adams M, Henderson MX. *West Michigan Regional Undergraduate Science Research Conference.* 2024

- **Gene Therapy in the Central Nervous System: ImageJ as a Tool for Image Analysis**

Nguyen YG. *West Michigan Regional Undergraduate Science Research Conference.* 2023

TEACHING EXPERIENCE

Teaching Assistant Mar - Jun 2024

Kalamazoo College, Kalamazoo, MI

- Graded and answered questions related to the assignments for the Applied Statistics course.

ADDITIONAL TRAINING

Neuromatch student in Computational Neuroscience course 2025

- Completed an intensive three-week, live program focused on practical applications of computational modeling and machine learning in neuroscience.

SERVICE AND OUTREACH

West Michigan Regional Undergraduate Science Research Conference 2026

Van Andel Institute, Grand Rapids, MI

- Supported an undergraduate research conference by welcoming attendees, assisting student poster presenters and recruiters, and helping coordinate on-site logistics.

SKILLS

Programming languages: R, Python, LaTeX

Analysis: statistical modeling, principal component analysis, clustering, nonlinear curve fitting, differential-expression analysis, spatial transcriptomics

Libraries: pandas, NumPy, Matplotlib, SuStaIn, Seurat, MAST, InSituType, ggplot2

Tools: Git, GitHub, Azure